Remark 3.15. By [19, Remark 2.28], if a definable function between two sets in Γ is a vrv-contraction then it is piecewise $\mathbb{Z}$-affine (with constant terms of the form $\mathrm{vrv}(t)$, where $t \in \mathrm{RV}$ is definable, that is, $t \in \mathrm{RV}(\mathbb{S})$). Moreover, a definable bijection between two objects of $\Gamma[k]$ is a $\Gamma[k]$-morphism if and only if it is definably a piecewise $\mathrm{GL}_k(\mathbb{Z})$-transformation. The "if" direction is clear. For the "only if" direction, see [8, Lemma 10.1] or [19, Lemma 2.29].

Resolved referee R050 [Clarification] – Piecewise integral-affine behavior.
Original source line 1289. The first sentence needs "piecewise $\mathbb{Z}$-affine," not globally "$\mathbb{Z}$-linear." For example, $\gamma \mapsto |\gamma|$ is the vrv-contraction of the definable map $t \mapsto t$ for $\mathrm{vrv}(t) \geq 0$, $t \mapsto t^{-1}$ otherwise. Its two integral-linear formulas differ. Retain the specified restriction on the translation constants.

Author resolution. The author approves the piecewise $\mathbb{Z}$-affine formulation. The restriction on translation constants and the following piecewise $\mathrm{GL}_k(\mathbb{Z})$ characterization are retained.

Definition 3.16 (Γ-categories with volume forms)**.** An object of the category $\mu\Gamma[k]$ is a definable pair (I, ω), where $I \in \Gamma[k]$ and $\omega : I \longrightarrow \Gamma$ is a function. A $\mu\Gamma[k]$-morphism between two objects (I, ω), (J, σ) is a $\Gamma[k]$-morphism $F : I \longrightarrow J$ such that for all $\alpha \in I$,

$$\omega(\alpha) = \sigma(F(\alpha)) + \mathrm{Jcb}_\Gamma F(\alpha).$$

The category $\mu\Gamma^{\mathrm{fin}}[k]$ is the obvious full subcategory of $\mu\Gamma[k]$.

Notation 3.17. For each $(U, f, \omega) \in \mu\mathrm{RV}[k]$, write $\omega_f : U \longrightarrow \Gamma$ for the function given by $u \longmapsto \Sigma(\mathrm{vrv} \circ f)(u) + \omega(u)$. Similarly, for $(I, \sigma) \in \mu\Gamma[k]$, write $\sigma_I : I \longrightarrow \Gamma$ for the function given by $\gamma \longmapsto \Sigma\gamma + \sigma(\gamma)$.

There is a natural map $\Gamma[*] \longrightarrow \mathrm{RV}[*]$ given by $I \longmapsto (I^\sharp, \mathrm{id})$ (see Notation 2.7). This map induces a commutative diagram in the category of graded semirings:

$$\begin{array}{ccc} \mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RES}[*] \\ \downarrow & & \downarrow \\ \mathbf{K}_+\,\Gamma[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RV}[*] \end{array}$$

where all the arrows are monomorphisms.

Proof of the horizontal monomorphisms. It suffices to prove that an $\mathbb{S}$-definable bijection $F : I^\sharp \longrightarrow J^\sharp$ in $\mathrm{RV}[n]$ implies that I and J are isomorphic in $\Gamma[n]$, retaining all finite disjoint-union tags. The case $n = 0$ is immediate, so suppose $n > 0$.

For each $\alpha \in I$, the full fibre $\alpha^\sharp$ is a torsor for $(\Bbbk^\times)^n$ and has a unique residue-generic type. By residue-field/value-group orthogonality, $\mathrm{vrv} \circ F$ has finite image on that fibre, with its target tag included. Exactly one target fibre $\beta^\sharp$ receives its generic part. Its image has residue dimension n, since F is injective, and is therefore generic in $\beta^\sharp$. The same argument for F^{-1} shows that this assignment defines a bijection $h : I \longrightarrow J$. Definability of residue dimension in families makes h definable over $\mathbb{S}$. After trivializing the two fibres solely for the proof, the generic restriction of F is a birational map of residue tori: in characteristic zero, the constructible maps supplied by F and F^{-1} are rational inverses on dense open subsets.

Fix $\alpha \in I$, put $\beta = h(\alpha)$, and set

$$\Delta = \mathrm{vrv}(\mathrm{RV} \cap \mathrm{dcl}(\mathbb{S})), \qquad L = (\Delta + \textstyle\sum_i \mathbb{Z}\alpha_i + \sum_i \mathbb{Z}\beta_i)/\Delta.$$

The group L is finitely generated. Let D be the diagonalizable group over $\Bbbk$ with character group L. Its two diagonal representations have weights $\bar{\alpha}_i$ and $\bar{\beta}_i$. Every character $\chi : L \longrightarrow \Bbbk^\times$ extends to Γ/Δ, since $\Bbbk^\times$ is divisible. The resulting RV twist $u \longmapsto \chi(\mathrm{vrv}(u))u$ fixes the residue field, every

Gamma value, and every old-ground RV point. It preserves the RV group operations and residue addition, so relative RV quantifier elimination makes it preserve every $\mathbb{S}$-definable RV relation, including the graph of F. The birational map of the two residue tori is thus D-equivariant. The two actions have the same kernel, by that map and its inverse, and their joint kernel is trivial because all their weights generate L. Both representations are faithful.

By [23, Theorem 7.1 and Corollary 2.5], there is a matrix $M \in \mathrm{GL}_n(\mathbb{Z})$ with $\beta - M\alpha \in \Delta^n$. Choose $c_i \in \mathrm{RV} \cap \mathrm{dcl}(\mathbb{S})$ with these values. Then

$$H(u)_i = c_i \prod_j u_j^{M_{ij}}$$

is an $\mathbb{S}$-definable bijection of the full fibres at α and β. This replaces the generic birational map; it need not extend the given map on its dense open domain. Compactness supplies finitely many such formulas on a definable partition of I. Their full-preimage maps glue to a bijection contracting to h, so h is a $\Gamma[n]$-morphism. This proves the lower horizontal positive injection. The upper injection follows from the commuting square and the two vertical positive injections. □

Author resolution (C002). The author approved adding the original-ground proof of the horizontal positive monomorphisms, using the indicated Reichstein–Youssin result. No HK theorem is taken as an input. There is no corresponding referee item.

The map

$$\mathbf{K}_+\,\mathrm{RES}[*] \times \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*]$$

determined by the assignment

$$([(U,f)],[I]) \longmapsto [(U \times I^\sharp, f \times \mathrm{id})] \tag{3.3}$$

is well-defined and is clearly $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-bilinear. Hence it induces a $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-linear map

$$\Psi : \mathbf{K}_+\,\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*], \tag{3.4}$$

which is a homomorphism of graded semirings. By the universal mapping property, groupifying a tensor product in the category of $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-semimodules is, up to isomorphism, the same as taking the corresponding tensor product in the category of $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$-modules; the groupification of Ψ is still denoted by Ψ. Similarly, there is a $\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]$-linear map

$$\mu\Psi : \mathbf{K}_+\,\mu\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\mu\Gamma[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}[*]. \tag{3.5}$$

We shall abbreviate $\otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]}$, $\otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]}$ both as "$\otimes$" below when no confusion can arise.

Proposition 3.18 (Positive tensor presentation)**.** *Let T_+ and $T_{\mu,+}$ be the sources of Ψ and $\mu\Psi$, respectively. All definitions below use the original ground $\mathbb{S}$. For a* RES *object with underlying set X, put*

$$\Lambda_X = \{\delta \in \Gamma \cap \mathrm{dcl}^{\mathrm{eq}}(\mathbb{S}) : \exists\ \mathbb{S}\text{-definable } c : X \longrightarrow \delta^\sharp\}.$$

Let $\mathcal{C}$ be the semiring congruence on T_+ generated by

$$[\boldsymbol{X}]_r \otimes [U]_m \ \sim\ [\boldsymbol{X}]_r \otimes [U + \delta e_i]_m, \qquad \boldsymbol{X} \in \mathrm{RES}[r],\ U \in \Gamma[m],\ \delta \in \Lambda_X,$$

where $m \geq 1$ and i is a coordinate index. Let $\mathcal{C}_\mu$ be the semiring congruence on $T_{\mu,+}$ generated by

$$[\boldsymbol{X}]_r \otimes [(U,\sigma)]_m \ \sim\ [\boldsymbol{X}]_r \otimes [(U + \delta e_i, \sigma_\delta)]_m, \qquad \sigma_\delta(\eta) = \sigma(\eta - \delta e_i) - \delta,$$

with $\boldsymbol{X} = (X, f, \omega) \in \mu\mathrm{RES}[r]$ and the same admissibility condition $\delta \in \Lambda_X$. The homomorphisms Ψ and $\mu\Psi$ are surjective, with kernel congruences $\mathcal{C}$ and $\mathcal{C}_\mu$, respectively. They induce isomorphisms of graded semirings

$$T_+/\mathcal{C} \ \simeq\ \mathbf{K}_+\,\mathrm{RV}[*], \qquad T_{\mu,+}/\mathcal{C}_\mu \ \simeq\ \mathbf{K}_+\,\mu\mathrm{RV}[*].$$